

Adding a Logging file for the EDA

In [30]:

```
import logging
from datetime import datetime

logging.basicConfig(
    filename="Aadhaar_Enrollment_Equity_Assessment_execution.log",
    filemode="a",
    level=logging.INFO,
    format="%(asctime)s | %(levelname)s | %(message)s"
)

logging.info("=====")
logging.info("EDA Phase-1 started")
logging.info(f"Notebook execution started at {datetime.now()}")
```

Importing Data From MySQL

In [31]:

```
import pandas as pd
from sqlalchemy import create_engine
engine = create_engine(
    "mysql+pymysql://root:PASSWORD%40629@localhost/uidai_clean"
)
uidai_clean = pd.read_sql(
    "SELECT * FROM uidai_district_enrollment_profile_clean",
    con=engine
)
```

In [32]:

```
uidai_clean.shape
uidai_clean.head()
```

Out[32]:

	state	district	Sum of total_enrolled	Sum of age_0_5	Sum of age_5_17	Sum of age_18_greater	Sum of child_ratio	Sum of youth_ratio	ad
0	Andaman & Nicobar Islands	Andamans	73	68	5	0	0.931507	0.068493	C
1	Andaman & Nicobar Islands	Nicobar	75	64	11	0	0.853333	0.146667	C
2	Andaman & Nicobar Islands	North And Middle Andaman	129	125	4	0	0.968992	0.031008	C
3	Andaman &	South Andaman	224	212	12	0	0.946429	0.053571	C

	state	district	Sum of total_enrolled	Sum of age_0_5	Sum of age_5_17	Sum of age_18_greater	Sum of child_ratio	Sum of youth_ratio	Sum of adult_ratio
	Nicobar Islands								
4	Andhra Pradesh	Alluri Sitharama Raju	1214	1064	116	34	0.876442	0.095552	0.028006

In [33]:

```
uidai_clean.columns = [
    "state",
    "district",
    "total_enrolled",
    "age_0_5",
    "age_5_17",
    "age_18_greater",
    "child_ratio",
    "youth_ratio",
    "adult_ratio",
    "dependency_ratio",
    "ESI",
    "InfiniteDependency"
]
```

In [34]:

```
uidai_clean.head()
```

Out[34]:

	state	district	total_enrolled	age_0_5	age_5_17	age_18_greater	child_ratio	youth_ratio	adult_ratio
0	Andaman & Nicobar Islands	Andamans	73	68	5	0	0.931507	0.068493	0.0
1	Andaman & Nicobar Islands	Nicobar	75	64	11	0	0.853333	0.146667	0.0
2	Andaman & Nicobar Islands	North And Middle Andaman	129	125	4	0	0.968992	0.031008	0.0
3	Andaman & Nicobar Islands	South Andaman	224	212	12	0	0.946429	0.053571	0.0
4	Andhra Pradesh	Alluri Sitharama Raju	1214	1064	116	34	0.876442	0.095552	0.028006

Which districts have structurally low Aadhaar enrollment maturity?

These districts need policy intervention, not cosmetic enrollment drives.

In [35]:

```
logging.info("Q1 started: Identifying structurally low enrollment maturity districts")

q1_low_maturity = uidai_clean[
    (uidai_clean["ESI"] < 0.45) &
    (uidai_clean["dependency_ratio"] > 1.2) &
    (uidai_clean["total_enrolled"] > 1000)
].sort_values("ESI")

q1_low_maturity[
    ["state", "district", "total_enrolled", "ESI", "dependency_ratio"]
].head(10)

logging.info("Q1 completed | districts_identified=%d", q1_low_maturity.shape[0])
display(q1_low_maturity.head(10))
```

	state	district	total_enrolled	age_0_5	age_5_17	age_18_greater	child_ratio	youth_rati
499	Odisha	Jharsuguda	1151	1066	85	0	0.926151	0.07384
631	Telangana	Komaram Bheem	1080	896	184	0	0.829630	0.17037
625	Telangana	Jangaon	1081	943	139	0	0.872340	0.12858
599	Tamil Nadu	Perambalur	2068	1842	226	0	0.890716	0.10928
605	Tamil Nadu	Tenkasi	1224	1111	113	0	0.907680	0.09232
597	Tamil Nadu	Nagapattinam	3935	3229	706	0	0.820584	0.17941
318	Karnataka	Tumkur	2895	2711	184	0	0.936442	0.06355
650	Telangana	Wanaparthy	1633	1242	391	0	0.760563	0.23943
246	Himachal Pradesh	Una	1203	1182	21	0	0.982544	0.01745
221	Haryana	Jind	3669	3579	90	0	0.975470	0.02453

Which states show high intra-state inequality in Aadhaar maturity?

This shows:

- The minimum ESI district
- The maximum ESI district
- The gap between the two shows the level of governance inequality

In [36]:

```
logging.info("Q2 started: Intra-state Aadhaar inequality with extreme districts")
```

```

uidai_esi = uidai_clean.loc[uidai_clean["ESI"].notna(), ["state", "district", "ESI"]]

esi_stats = (
    uidai_esi
    .groupby("state", as_index=False)
    .agg(
        districts=("district", "nunique"),
        min_ESI=("ESI", "min"),
        max_ESI=("ESI", "max")
    )
)

min_districts = (
    uidai_esi
    .loc[uidai_esi.groupby("state")["ESI"].idxmin()]
    .rename(columns={
        "district": "min_ESI_district",
        "ESI": "min_ESI"
    })
)

max_districts = (
    uidai_esi
    .loc[uidai_esi.groupby("state")["ESI"].idxmax()]
    .rename(columns={
        "district": "max_ESI_district",
        "ESI": "max_ESI"
    })
)

q2 = (
    esi_stats
    .merge(min_districts[["state", "min_ESI_district"]], on="state", how="left")
    .merge(max_districts[["state", "max_ESI_district"]], on="state", how="left")
)

q2["ESI_gap"] = q2["max_ESI"] - q2["min_ESI"]

display(
    q2.sort_values("ESI_gap", ascending=False).head(10)
)

logging.info("Q2 completed | states=%d", q2.shape[0])

```

	state	districts	min_ESI	max_ESI	min_ESI_district	max_ESI_district	ESI_gap
22	Meghalaya	13	0.000000	2.572052	West Jaintia Hills	Eastern West Khasi Hills	2.572052
23	Mizoram	11	0.000000	1.833333	Hnahthial	Khawzawl	1.833333
2	Arunachal Pradesh	25	0.000000	0.500000	Dibang Valley	Leparada	0.500000

	state	districts	min_ESI	max_ESI	min_ESI_district	max_ESI_district	ESI_gap
27	Punjab	23	0.002869	0.483078	Ferozepur	Kapurthala	0.480210
35	West Bengal	21	0.002437	0.435897	Murshidabad	Kalimpong	0.433461
24	Nagaland	17	0.039039	0.450000	Phek	Tseminyu	0.410961
19	Madhya Pradesh	55	0.000382	0.400000	Dindori	Pandhurna	0.399618
3	Assam	35	0.000000	0.368421	Tamulpur District	West Karbi Anglong	0.368421
6	Chhattisgarh	33	0.000000	0.255170	Khairagarh Chhuikhadan Gandai	Vijayapura	0.255170
28	Rajasthan	35	0.000000	0.210526	Didwana-Kuchaman	Deeg	0.210526

Meghalaya is the state with the largest ESI gap

Which districts are child-heavy enrollment zones?

These districts indicate future UIDAI workload and school-linked identity needs.

Child-heavy districts:

child_ratio > 0.35

total_enrolled > 1000

In [37]:

```
from IPython.display import display, Markdown

logging.info("Q3 started: Identifying child-heavy enrollment districts")

q3_child_heavy = uidai_clean.loc[
    (uidai_clean["child_ratio"].notna()) &
    (uidai_clean["child_ratio"] > 0.35) &
    (uidai_clean["total_enrolled"] > 1000),
    ["state", "district", "child_ratio", "total_enrolled"]
].sort_values("child_ratio", ascending=False)

display(Markdown("### Top 10 Child-Heavy Districts Across India"))
display(q3_child_heavy.head(10))

q3_top_by_state = (
    q3_child_heavy
    .loc[q3_child_heavy.groupby("state")["child_ratio"].idxmax()]
    .sort_values("child_ratio", ascending=False)
)

display(Markdown("### Most Child-Heavy District in Each State"))
display(q3_top_by_state)

display(Markdown("### Complete List of Child-Heavy Districts"))
display(q3_child_heavy)
```

```
logging.info(
    "Q3 completed | districts_identified=%d | states_covered=%d",
    q3_child_heavy.shape[0],
    q3_top_by_state["state"].nunique()
)
```

Top 10 Child-Heavy Districts Across India

	state	district	child_ratio	total_enrolled
237	Himachal Pradesh	Hamirpur	0.986195	1159
246	Himachal Pradesh	Una	0.982544	1203
242	Himachal Pradesh	Mandi	0.978243	2436
238	Himachal Pradesh	Kangra	0.975572	3848
221	Haryana	Jind	0.975470	3669
401	Maharashtra	Bhandara	0.967240	1862
25	Andhra Pradesh	Srikakulam	0.967193	4176
222	Haryana	Kaithal	0.966048	2857
234	Haryana	Yamunanagar	0.964046	3282
224	Haryana	Kurukshetra	0.963235	2312

Most Child-Heavy District in Each State

	state	district	child_ratio	total_enrolled
237	Himachal Pradesh	Hamirpur	0.986195	1159
221	Haryana	Jind	0.975470	3669
401	Maharashtra	Bhandara	0.967240	1862
25	Andhra Pradesh	Srikakulam	0.967193	4176
130	Chhattisgarh	Balod	0.956767	1596
311	Karnataka	Mandya	0.945839	3545
368	Madhya Pradesh	Mandla	0.945668	4859
592	Tamil Nadu	Kanniyakumari	0.943015	4159
624	Telangana	Jagitial	0.932389	1553
259	Jammu & Kashmir	Pulwama	0.928415	1830
499	Odisha	Jharsuguda	0.926151	1151
535	Punjab	Pathankot	0.925590	1102
518	Puducherry	Puducherry	0.922948	2388
739	Uttarakhand	Chamoli	0.919617	1045
129	Chandigarh	Chandigarh	0.907146	2617
201	Gujarat	Narmada	0.900485	1236
766	West Bengal	Murshidabad	0.874886	34968

	state	district	child_ratio	total_enrolled
337	Kerala	Wayanad	0.833885	1511
163	Dadra And Nagar Haveli And Daman And Diu	Dadra And Nagar Haveli	0.832056	1566
178	Goa	North Goa	0.818939	1320
108	Bihar	Kishanganj	0.793074	8230
171	Delhi	North West Delhi	0.771305	15501
679	Uttar Pradesh	Bijnor	0.765643	16445
559	Rajasthan	Ganganagar	0.762411	5358
284	Jharkhand	Palamu	0.761093	8677
72	Assam	Hojai	0.750432	6940
338	Ladakh	Kargil	0.736166	1012
458	Mizoram	Aizawl	0.732478	1013
660	Tripura	West Tripura	0.701863	2737
437	Manipur	Imphal West	0.520953	1217
452	Meghalaya	South West Garo Hills	0.487637	4934
479	Nagaland	Peren	0.350793	1451

Complete List of Child-Heavy Districts

	state	district	child_ratio	total_enrolled
237	Himachal Pradesh	Hamirpur	0.986195	1159
246	Himachal Pradesh	Una	0.982544	1203
242	Himachal Pradesh	Mandi	0.978243	2436
238	Himachal Pradesh	Kangra	0.975572	3848
221	Haryana	Jind	0.975470	3669
...	...	...	...	...
75	Assam	Karbi Anglong	0.357726	7967
128	Bihar	West Champaran	0.355147	46271
731	Uttar Pradesh	Siddharthnagar	0.353214	18125
479	Nagaland	Peren	0.350793	1451
730	Uttar Pradesh	Shravasti	0.350059	7573

613 rows × 4 columns

- **Hamirpur is the most child-heavy district.**
- **Himachal Pradesh, Haryana and Maharashtra are the three most child-heavy states of India.**

In [38]:

```

logging.info("Q4 started: Near-saturation Aadhaar districts")

q4 = uidai_clean.loc[
    (uidai_clean["ESI"].notna()) &
    (uidai_clean["dependency_ratio"].notna()) &
    (uidai_clean["ESI"] > 0.75) &
    (uidai_clean["dependency_ratio"] < 0.6)
]

q4_count = q4.shape[0]

display(pd.DataFrame({"near_saturation_districts": [q4_count]}))

logging.info("Q4 completed | near_saturation_districts=%d", q4_count)
display(q4)

```

near_saturation_districts	
0	2

	state	district	total_enrolled	age_0_5	age_5_17	age_18_greater	child_ratio	youth_ratio
448	Meghalaya	Eastern West Khasi Hills	818	3	226	589	0.003667	0.276284
461	Mizoram	Khawzawl	34	5	7	22	0.147059	0.205882

- Only 2 states of India have districts which are near saturation.
- This indicates that Aadhaar enrollment remains a structurally continuous process, driven by India's demographic depth rather than administrative lag.

Number of districts in each maturity stage

In [39]:

```

import numpy as np
logging.info("Q5 started: Enrollment maturity classification")

uidai_q5 = uidai_clean.copy()

uidai_q5["maturity_stage"] = np.select(
    [
        uidai_q5["ESI"] < 0.45,
        uidai_q5["ESI"].between(0.45, 0.65),
        uidai_q5["ESI"] > 0.65
    ],
    [
        "Emerging",
        "Transitional",
        "Mature"
    ],
    default="Unknown"
)

```



```

)

q5 = (
    uidai_q5["maturity_stage"]
    .value_counts()
    .rename_axis("Maturity Stage")
    .reset_index(name="Number of Districts")
)

display(q5)

logging.info("Q5 completed")

```

	Maturity Stage	Number of Districts
0	Emerging	759
1	Transitional	10
2	Mature	3

This shows most of the districts in the country are from the emerging maturity stage where the $ESI < 0.45$ that is less than 45% enrollments in the districts are adults.

Districts with Administrative/Data Anomalies

In [40]:

```

logging.info("Q6 started: Administrative anomaly detection")

q6 = uidai_clean.loc[
    (uidai_clean["total_enrolled"] < 500) |
    (uidai_clean["InfiniteDependency"] == 1) |
    (uidai_clean["ESI"].isna()),
    [
        "state",
        "district",
        "total_enrolled",
        "ESI",
        "dependency_ratio",
        "InfiniteDependency"
    ]
]

display(q6.head(10))

logging.info("Q6 completed | rows=%d", q6.shape[0])

```

	state	district	total_enrolled	ESI	dependency_ratio	InfiniteDependency
0	Andaman & Nicobar Islands	Andamans	73	0.000000	50.000000	1
1	Andaman & Nicobar Islands	Nicobar	75	0.000000	50.000000	1
2	Andaman & Nicobar Islands	North And Middle Andaman	129	0.000000	50.000000	1

	state	district	total_enrolled	ESI	dependency_ratio	InfiniteDependency
3	Andaman & Nicobar Islands	South Andaman	224	0.000000	50.000000	1
8	Andhra Pradesh	Bapatla	482	0.034335	29.125000	0
21	Andhra Pradesh	Parvathipuram Manyam	489	0.016632	60.125000	0
31	Arunachal Pradesh	Anjaw	36	0.121212	8.250000	0
33	Arunachal Pradesh	Dibang Valley	10	0.000000	206.000000	1
34	Arunachal Pradesh	East Kameng	179	0.084848	11.785714	0
35	Arunachal Pradesh	East Siang	165	0.012270	81.500000	0

These districts show unusual structural enrollment behaviour, not because of incorrect data, but due to specific demographic conditions present in the dataset.

These anomalies can arise because of:

- Total enrollments < 500, representing low Aadhaar enrollment activity or sparse administrative participation.
- Infinite dependency conditions caused by zero adult enrollments (`age_18_greater = 0`). This does not indicate wrong data; rather, it reflects districts where no adult population was recorded in the enrollment structure.
- ESI = 0, indicating absence of adult enrollment contribution within the demographic composition.

To preserve structural comparability during analysis, districts with infinite dependency were not removed. Instead, their dependency ratio was dynamically reassigned using the following logic:

Dependency Ratio Adjustment Logic

Modified Dependency Ratio = (Maximum Dependency Ratio of the Corresponding State + 50)

This approach ensures that:

- Infinite dependency districts remain analytically distinguishable
- State-level structural imbalance is still reflected
- The demographic instability analysis remains meaningful without introducing undefined values

In []: